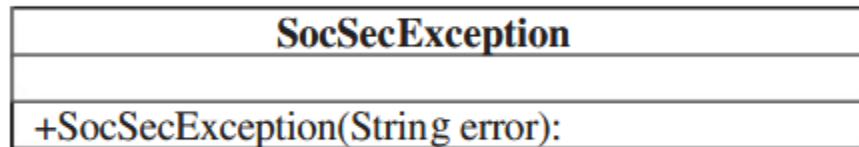
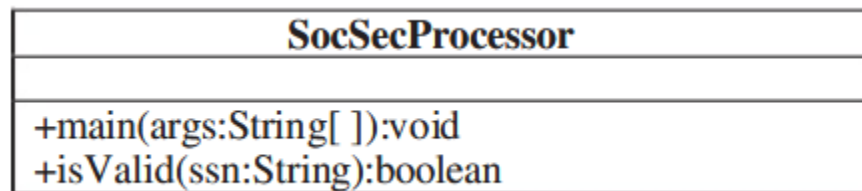


Task #1 Writing a Custom Exception Class

1. Create an exception class called SocSecException. The UML for this class is below. The constructor will call the superclass constructor. It will set the message associated with the exception to “Invalid social security number” concatenated with the error string.



2. Create a driver program called SocSecProcessor.java. This program will have a main method and a static method called isValid that will check if the social security number is valid.



Task #2 Writing Code to Handle an Exception

1. In the main method:
 - a) The main method should read a name and social security number from the user as Strings.
 - b) The main method should contain a try-catch statement. This statement tries to check if the social security number is valid by using the method isValid. If the social security number is valid, it prints the name and social security number. If a SocSecException is thrown, it should catch it and print out the name, social security number entered, and an associated error message indicating why the social security number is invalid.
 - c) A loop should be used to allow the user to continue until the user indicates that they do not want to continue.
2. The static isValid method:
 - a) This method throws a SocSecException.
 - b) True is returned if the social security number is valid, false otherwise.
 - c) The method checks for the following errors and throws a SocSecException with the appropriate message.
 - i. Number of characters not equal to 11. (Just check the length of the string)
 - ii. Dashes in the wrong spots.

- iii. Any non-digits in the SSN.
 - iv. Hint: Use a loop to step through each character of the string, checking for a digit or hyphen in the appropriate spots.
3. Compile, debug, and run your program. Sample output is shown below with user input in bold.

OUTPUT (boldface is user input)

Name? **Sam Sly**
SSN? **333-00-999**
Invalid the social security number, wrong number of characters
Continue? **y**
Name? **George Washington**
SSN? **123-45-6789**
George Washington 123-45-6789 is valid
Continue? **y**
Name? **Dudley Dorigt**
SSN? **222-00-999o**
Invalid the social security number, contains a character that is not a digit
Continue? **Y**
Name? **Jane Doe**
SSN? **333-333-333**
Invalid the social security number, dashes at wrong positions
Continue? **n**